THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student wants to know if the amount of sunlight affects growth of bean seedlings. Which should be the same for all pots? 1

(a) amount of sunlight **(b)** amount of water and type of soil
(c) height at the end **(d)** number of leaves at the end

2. Why is an experiment repeated several times? 1

(a) to make sure the result is reliable **(b)** to waste time
(c) to change the question **(d)** to avoid recording data

3. The SI unit of time used in a stopwatch reading is: 1

(a) kelvin **(b)** second **(c)** kilogram **(d)** metre

4. Which is an example of a prediction? 1

(a) The puri puffed in 6 s. **(b)** The dough is soft.
(c) The puri in hotter oil will puff faster. **(d)** The oil is hot.

5. Which instrument is best to measure the time taken for a puri to puff up? 1

(a) spring balance **(b)** stopwatch **(c)** thermometer **(d)** metre scale

6. Different observers measuring the same length with the same scale get slightly different values. This difference suggests: 1

(a) length does not exist **(b)** measurements have some uncertainty
(c) the scale is magical **(d)** science is useless

7. Science progresses because scientists: 1

(a) only trust authority
(b) share, question and verify each other's work
(c) never test old ideas
(d) keep results secret forever

8. The statement drawn from the observations at the end of an investigation is the: 1

(a) question **(b)** apparatus **(c)** variable **(d)** conclusion

9. Taking the average of several readings is done to: 1

(a) increase the error **(b)** avoid repeating
(c) get a more dependable value **(d)** hide odd values

10. An unexpected result in an experiment should be: 1

(a) ignored **(b)** replaced by the expected result
(c) recorded and investigated further **(d)** deleted

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a table useful in recording observations? 2

12. Differentiate between a qualitative and a quantitative observation with one example each. 2

13. Why is the scientific method called flexible? 2

14. Give two reasons why a question like 'Which puri is the most beautiful?' is not suitable for scientific investigation. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A student's data show that puris puff in 10 s, 9 s and 11 s at the same temperature. Find the average and say why it is better to quote it. 3

16. State three precautions to be taken while using a thermometer to measure the temperature of hot oil. 3

17. Distinguish between a hypothesis and a conclusion. 3

OR

Explain how a graph of time taken versus oil temperature helps to draw a conclusion. 3

18. Give three examples of how science is useful in everyday kitchen work. 3

OR

List three steps in recording and presenting data from an investigation. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A student kept three bean seeds in three cups: dry cotton, wet cotton and water covering the seeds. Name the variable, identify what stays constant, and say which cup would show germination best. 5

OR

A graph of puffing time against oil temperature slopes downward. Describe what it shows and one limitation of drawing a conclusion from only a few points. 5

20. Explain the importance of honesty, curiosity and teamwork in science with examples. 5

OR

Discuss how observation and curiosity led to scientific ideas, using any two everyday situations. 5

END OF QUESTION PAPER Total: 40 Marks